

Cambridge Lower Secondary Checkpoint

MATHEMATICS**0862/02**

Paper 2

October 2025

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Markers were instructed to award marks. It does not indicate the details of the discussions that took place at a Markers' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the End of Series Report. Cambridge will not enter into discussions about these mark schemes.

This document has **12** pages. Any blank pages are indicated.

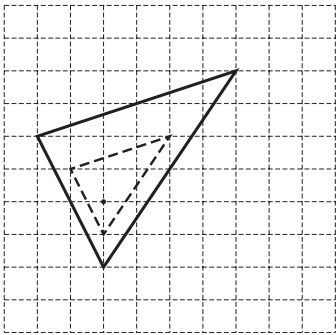
Mark scheme annotations and abbreviations

FT	follow through after error
SC	special case mark
cao	correct answer only
dep	dependent
isw	ignore subsequent working
nfww	not from wrong working
oe	or equivalent
soi	seen or implied

Question	Answer	Marks	Part marks	Guidance
1(a)	2π	1		Accept any clear indication.
1(b)	8	1		Accept ± 8 or -8
2	3×10^4	1		
3	☒ ☐ ☐ ☒	1		Both answers correct for the mark. Accept any clear indication.
4	True False ☒ ☐ ☐ ☒	1		Both answers correct for the mark. Accept any clear indication.
5	-4	1		
6	$(14 + \sqrt{36}) \div 2 = 10$	1		More than one pair of brackets scores 0
7	$x = -3$	1		
8	$(n + 3)^2$ oe	1		Oe, e.g. $n^2 + 6n + 9$ Do not accept equations, e.g. $n = (n + 3)^2$, $x = (n + 3)^2$
9	14 36	2	Award 1 mark for 14 or 36	Correct order only.
10	17 or 18 or 19 or 20	1		

Question	Answer	Marks	Part marks	Guidance
11(a)	6	1		
11(b)	n^3 oe	1		Accept unsimplified, e.g. $n \times n \times n$.
12	[Boat] × positioned on a bearing of 110° and a distance of 8.5 cm from L .	2	Award 1 mark for bearing of 110° drawn or for × being a distance of 8.5 cm from L .	Tolerance for all marks is $\pm 2^\circ$ and ± 2 mm. Do not accept the end of a line alone to represent the position of the boat.
13(a)	72°	1		
13(b)	36	2	Award 1 mark for $(180 - 170) \times n = 360$ oe or $\frac{180(n-2)}{n} = 170$ oe.	Oe, e.g. $\frac{360}{180-170}$ Oe, e.g. $180n - 360 = 170n$
14(a)	0.17	1		Accept equivalent fractions or percentage.
14(b)	228	2	Award 1 mark for $0.2 + 0.18$ oe or for 120 or 108	Implied by 0.38, $\frac{19}{50}$, 38% Award 1 mark for answer $\frac{228}{600}$

Question	Answer		Marks	Part marks	Guidance
15	Regular	Not regular	1		All three ticks correct for the mark. Accept any clear indication.
		✓			
	✓				
		✓			
16	3.15 (≤L<) 3.25		2	Award 1 mark for one correct or for both correct but reversed.	

Question	Answer	Marks	Part marks	Guidance
17	Correct image drawn, i.e. 	2	Award 1 mark for two vertices of image correctly positioned or if image has the correct size and shape but incorrectly positioned on the grid.	Must be on given grid for this mark.
18	$(x =) -4$ and $(y =) 5$	3	Award 2 marks for $(x =) -4$ or $(y =) 5$ or Award 1 mark for a correct method to eliminate one variable or for <i>their</i> two values satisfying one of the original equations.	Correct method, e.g. Multiplying one or both equations by a constant to get equal coefficients and correct consistent addition or subtraction, implied by $4y = 20$ or making x or y the subject and substituting into the other equation, e.g. $3(16 - 4y) + 8y = 28$, $\frac{28 - 8y}{3} + 4y = 16$

Question	Answer	Marks	Part marks	Guidance
19	1270 or 1271 to 1273 (cm ³)	2	Award 1 mark for $\pi \times 4.5^2$ [$\times 20$] or better.	For 2 marks, accept 405π . Or better, e.g. 63.5 to 63.6..., $\frac{81\pi}{4}$, 20.25π . Accept values of π between 3.14 and $\frac{22}{7}$ in the method.

Question	Answer	Marks	Part marks	Guidance
20	$(x =) [\pm] \sqrt{\frac{y+a}{6}}$ oe	3	Award 1 mark for correctly isolating term in x^2 and Award 1 mark for correctly isolating x^2 FT <i>their</i> previous step and Award 1 mark for correctly square rooting FT <i>their</i> previous step to a maximum of 2 marks for an incorrect answer.	e.g. $y + a = 6x^2$ e.g. $\frac{y+a}{6} = x^2$ Square root sign must extend beyond the operation sign in the numerator and below the fraction line. Note steps will vary depending on starting point and can be in any order. Always FT a correct step after <i>their</i> previous completed step, e.g. $\frac{y}{6} = x^2 - \frac{a}{6}, \frac{y}{6} + \frac{a}{6} = x^2, \sqrt{\frac{y}{6} + \frac{a}{6}} = x$ $y + a = 6x^2, \sqrt{y+a} = \sqrt{6}x, \frac{\sqrt{y+a}}{\sqrt{6}} = x$ $-y - a = -6x^2, \frac{-y-a}{-6} = x^2, \sqrt{\frac{-y-a}{-6}} = x$ (Note the final alternative equivalent answers each scoring 3 marks in these methods).
21	207 000 cao 0.0348 cao	2	Award 1 mark for each correct answer.	Correct order only. Do not accept 0.03480
22	100 3 -3	3	Award 1 mark for each correct value.	Accept 3 and -3 in either order.

Question	Answer	Marks	Part marks	Guidance
23	$(y =) -\frac{3}{2}x + 1$	2	Award 1 mark for [gradient =] $-\frac{3}{2}$ oe or $(y =) mx + 1$ where $m \neq 0$	Accept equivalents for $-\frac{3}{2}$ for 1 or 2 marks. 1 mark implied by, e.g. $\left[\frac{4 - -2}{-2 - 2} = \right] \frac{6}{-4}$ $[m =] -1.5, \frac{-3}{2}x + c, c$ can be 0 m can be left as the letter m for $y = mx + 1$

Question	Answer	Marks	Part marks	Guidance
24	111 or 110.5 to 110.6 (cm ²)	4	Award 3 marks for $\frac{1}{2} \times \pi \times \left(\frac{12}{2}\right)^2 + \frac{1}{2} \times 12 \times \sqrt{15^2 - 12^2} \text{ oe}$ or Award 2 marks for $\frac{1}{2} \times \pi \times \left(\frac{12}{2}\right)^2 \text{ oe}$ or $\sqrt{15^2 - 12^2} \text{ oe}$ or Award 1 mark for $\pi \times \left(\frac{12}{2}\right)^2 \text{ oe}$ or for $x^2 + 12^2 = 15^2 \text{ oe}$ or better.	Accept answer $18\pi + 54$ for 4 marks. Oe, e.g. $\frac{1}{2} \times \pi \times 6^2 + \frac{1}{2} \times 12 \times 9$ 2 marks implied by 18π or 56.5 to 56.6 or by $AC = 9$, area of triangle = 54 Oe, e.g. $\sqrt{225 - 144} \sqrt{81}$, 1 mark implied by 36π or 113[.0] to 113.2 Or better, e.g. $15^2 - 12^2$ or 81 Marks may be awarded on diagram. For 2 marks or 1 mark you should mark to the candidate's advantage, e.g. do not withdraw marks for correct Pythagoras work if values found are not used or not used correctly in subsequent area work. Accept values of π between 3.14 and $\frac{22}{7}$

Question	Answer	Marks	Part marks	Guidance
25	1.58 or 1.576... (kg)	4	Award 3 marks for $\frac{7 \times 1.3 + 10 \times 1.5 + 14 \times 1.7 + 3 \times 1.9}{34}$ or better or Award 2 marks for $7 \times 1.3 + 10 \times 1.5 + 14 \times 1.7 + 3 \times 1.9$ or better or Award 1 mark for either of these: three correct midpoints out of 1.3, 1.5, 1.7 and 1.9 three correct frequencies out of 7, 10, 14 and 3	Accept $7 + 10 + 14 + 3$ in place of 34 For 3 marks, accept one error in one of the products implied by three values used from 9.1, 15, 23.8, 5.7 Or better, e.g. $\frac{9.1 + 15 + 23.8 + 5.7}{34}$, $\frac{53.6}{34}$ For 2 marks, accept one error in one of the products and accept midpoints as consistent values within intervals including either end of intervals. Or better, e.g. $9.1 + 15 + 23.8 + 5.7$, 53.6 Using start of intervals: 2 marks implied by $8.4 + 14 + 22.4 + 5.4$, 50.2, answer 1.48 or 1.476... Using end of intervals: 2 marks implied by $9.8 + 16 + 25.2 + 6$, 57, answer 1.68 or 1.676... May be seen on diagram. 1 mark implied by three values correct from 9.1, 15, 23.8, 5.7 (or using consistent values within intervals). Is w interval stated after correct mean found.

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